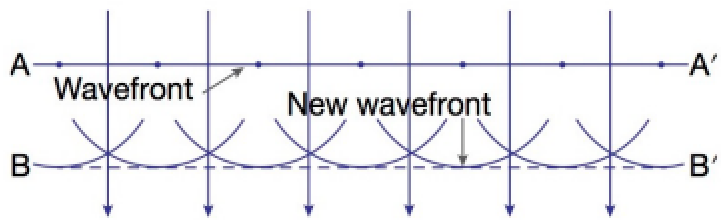
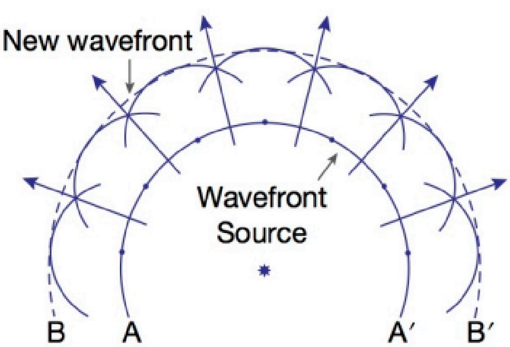
The Nature of Light

Huygen and Newton's Model of Light

- Up to the late 17th century, there was debate about the true nature of light.
- In 1690, Christiaan Huygens published 'Treatise on Light' outlining his wave theory of light.
    - Huygens stated that each point on the wavefront acts as a fresh source of distribution of light and all points on an initial wavefront act as sources of spherical wavelets.



- Huygens also stated that secondary wavelets radiate out, forming a new resultant wavefront.



- Huygens' theory explained the behaviour of waves, such as propagation, reflection and refraction, diffraction, interference and polarisation.

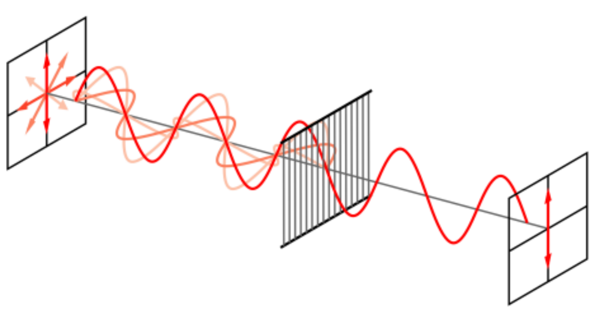
- Newton published 'Opticks' in 1704, outlining his corpuscular model of light.

- He stated that a light source emits light particles called corpuscles, which travel through transparent media in straight lines.
- He also stated that corpuscles enter the eyes to produce the sensation of vision, and different colours of light are due to different sized corpuscles.

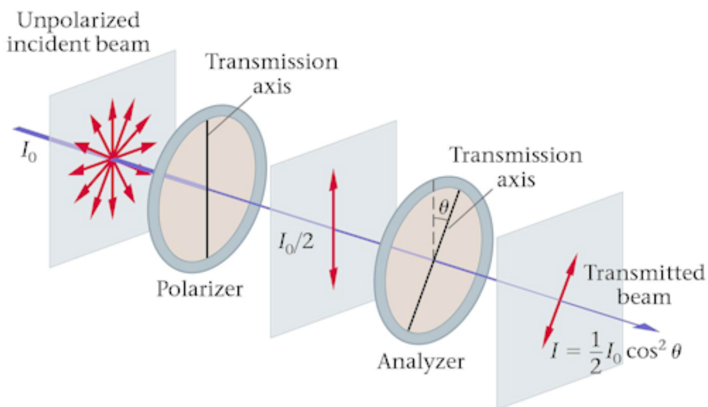
- Newton's corpuscular model requires certain properties for the corpuscles:
  - They are perfectly elastic and rigid, with negligible mass.
  - They are repelled by a reflecting surface to explain reflection.
  - They are attracted by transparent materials to explain travelling in straight lines and refraction.
- Newton's model was limited because it could not explain simultaneous reflection and refraction, and could not explain diffraction, interference and polarisation.
- Newton's model was favoured due to his status and influence within the scientific community, as well as because Huygen's wave model had limited experimental evidence.

Polarisation

- Polarisation refers to the orientation of the oscillation of a transverse wave.
  - It was first observed by Rasmus Bartholin in 1669 and first explained by Huygens' wave model of light.
  - It was explained by Newton by considering elongated corpuscles.
- A polariser is an optical filter that only transmits components of light waves with a specific orientation.
  - This means that components in a different orientation are blocked.
  - These polarisers can be made of tourmaline, herapathite, iodine-doped PVA, or similar materials.



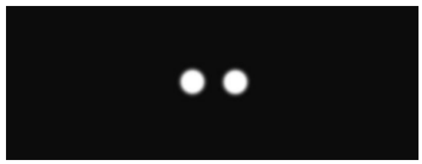
- Etienne-Louis Malus proposed a law in 1808 that calculated the intensity of light from two polarisers.
    - This is expressed in the formula $I = I_{max} cos^2\theta$, where I_{max} is the original intensity of the light and θ is the angle between the two orientations of light.



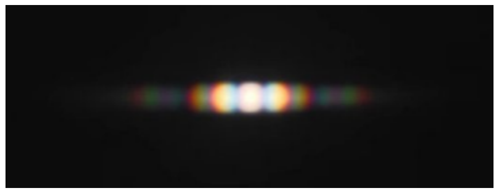
- The intensity of light is halved when it first goes through a polariser.
- Malus' Law explains why parallel polarisers allow light to go through, while perpendicular polarisers block all light.
    - A rotated intermediate polariser between perpendicular polarisers allows light to go through.
    - This is because it is two applications of $\theta = 45°$, rather than $\theta = 90°$.

Young's Double-Slit Experiment

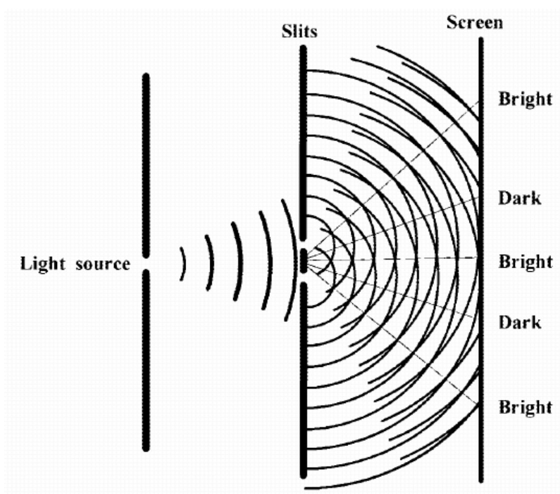
- In 1803, Thomas Young reported on an experiment where sunlight was passed through a collimating slit, a thin card being placed in front of the beam and the split light projected onto a screen.
- Newton's particle model hypothesised that there would only be two spots of full intensity projected onto the screen.



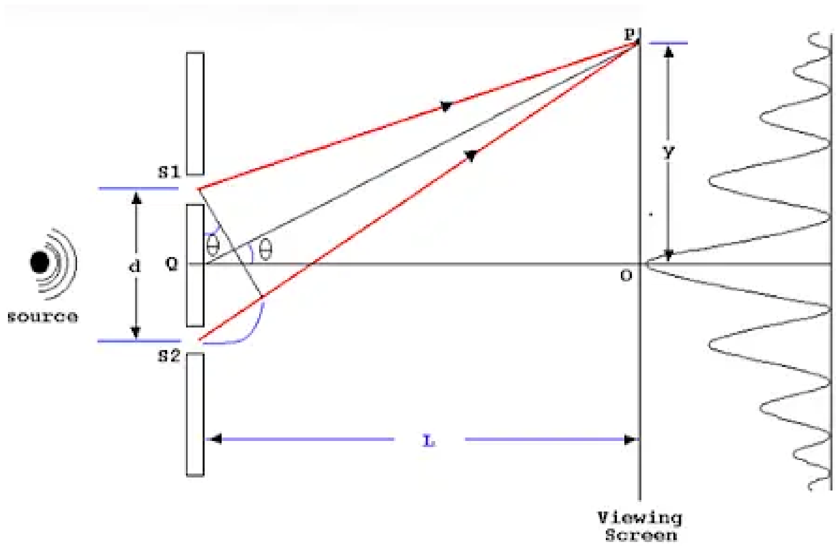
- However, it was observed that there were many spots of varying intensities projected.
    - These spots were most intense in the middle, and became less bright the further they were from the collimating strip.
    - These observations aligned with Huygens' wave model.



- There were alternating light and dark regions from the constructive and destructive interference of light, as the light diffracted around the collimating strip.



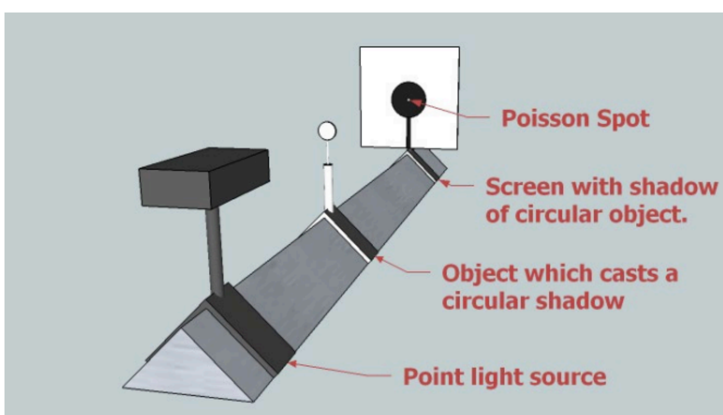
- The intensity profile of the pattern depends on the size of the slits, the separation of the slits, and the wavelength of light.
- For calculating the angle of constructive interference at the screen, the formula $d sin\theta = m\lambda$ is used.
    - m refers to the number of the bright spot, so for calculating the first-order bright spot, $m = 1$.



- Using the small angle approximation of $sin\,\theta \approx tan\,\theta \approx \theta$, for $\theta \ll 1$ rad, rearranging for y gives $y \approx \frac{m\lambda L}{d}$.

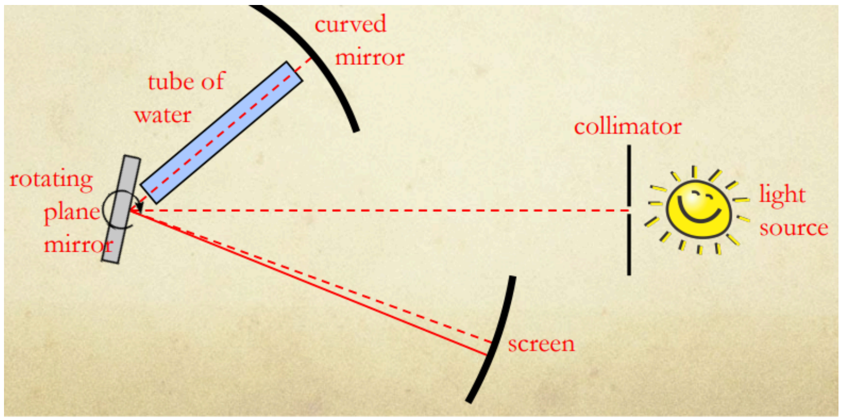
Arago Spot and Foucault's Experiment

- In 1818, the French Academy of Sciences held a competition to explain the properties of light.
  - In this competition, Augustin-Jean Fresnel submitted a new wave theory of light.
- Fresnel's model of light meant that a bright spot would appear at the centre of a circular shadow, which Simeon Denis Poisson mentioned.
  - Francois Arago, the head of the judging committee, carried out an experiment to test Poisson's argument in 1819.



- The appearance of the Poisson spot, or the Arago spot, convinced most scientists of the wave nature of light.

- Leon Foucault analysed the speed of light in 1850, and added a tube of water between mirrors to see if a different medium would slow it down.



- The tube of water had slowed the speed of light, which was a contradiction of Newton's explanation for refraction.
    - This ended the corpuscular model of light.

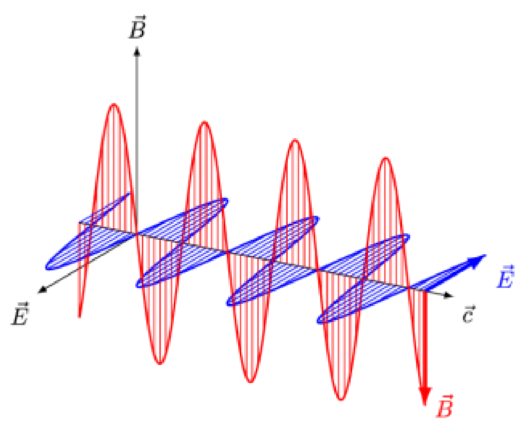
Electromagnetic Model

Maxwell's Contributions

- In 1820, Hans Christian Ørsted discovered that a current produces a magnetic field, and in 1831, Michael Faraday discovered that a changing magnetic field induces a current.
    - James Maxwell combined electricity and magnetism in 1855-56 to start the field of electromagnetism.
- Maxwell's notable pieces of work are as follows:
    - 'On Faraday's Lines of Force' (1855) mathematically developed Faraday's concept of a field, or lines of force.

- ‘On Physical Lines of Force’ (1861) derived the Maxwell equations of electromagnetism.
- ‘A Dynamical Theory of the Electromagnetic Field’ (1865) derived an equation for the speed of electromagnetic waves.

- Maxwell’s model of electromagnetism was made up of four Maxwell equations:
  - Gauss’ law ($\nabla \cdot E = \frac{\rho}{\varepsilon 0}$) describes the electric flux produced by electric charges, where ρ is the total electric charge density and ε_0 is the electric permittivity constant.
  - Gauss’ law for magnetism ($\nabla \cdot B = 0$) states that the magnetic flux through any closed surface is equal to zero.
  - Faraday’s law of induction ($\nabla \times E =- \frac{\delta B}{\delta t}$) describes the electric field induced by a changing magnetic field.
  - Ampere’s circuit law ($\nabla \times B = \mu_0 j + \frac{1}{c^2}\frac{\delta E}{\delta t}$) describes the magnetic field produced by an electric current or by a changing electric flux, where c is the speed of electromagnetic waves and μ_0 is the magnetic permeability constant.
- Maxwell reasoned a mutual induction of changing electric and magnetic fields, which was able to sustain itself.

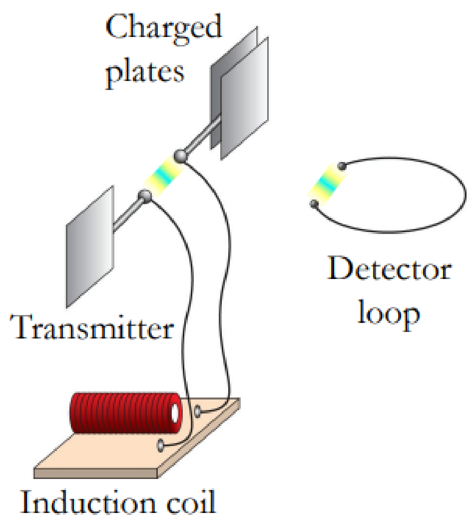


- Maxwell’s equations allowed him to calculate the speed of this wave, which was given as $c = \frac{1}{\sqrt{\mu 0 \varepsilon 0}}$.

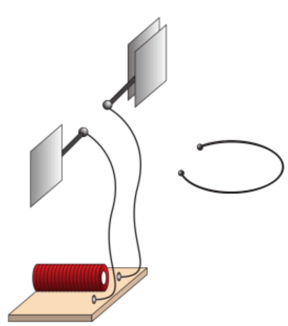
    - Using values available at the time, it was calculated $c = 3.1074 \times 10^8$ ms^{-1}.
- Maxwell observed that this value was very close to measured values for the speed of light.
    - He surmised in 1865 that "*light and magnetism are affections of the same substance*", and that "*light is an electromagnetic disturbance propagated through the field according to electromagnetic laws*".
- Maxwell also predicted the existence of different types of EMR that was not visible.
    - This was confirmed in 1800 when William Herschel discovered infrared radiation and in 1801 when Johann Ritter discovered ultraviolet radiation.
    - Neither scientist realised their discovery was a form of EMR.

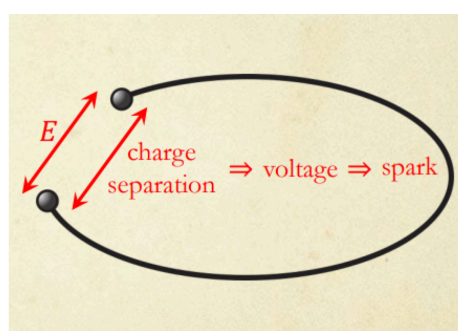
Confirmation of Maxwell and Polarisation of EMR

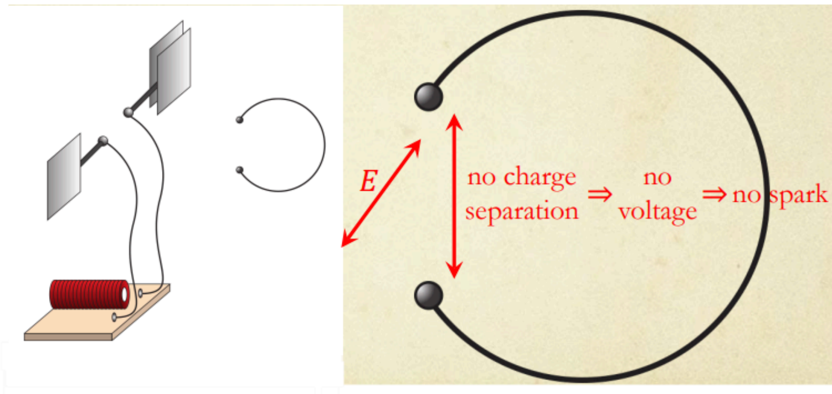
- A stationary charge produces an electric field and a moving charge produces a magnetic field.
- An accelerating charge produces EMR.
    - This includes electrons moving in antennas, ions moving in plasmas in stars and electricity jumping across gaps.
- In 1886, Heinrich Hertz tried to verify Maxwell's predictions.
    - He used an induction coil to produce sparks across a gap between two spherical electrodes, and observed sparks across a similar gap in a loop of wire nearby.



- The sequence that occurred in the experiment is below:
    1. The induction coil produces oscillating high voltage.
    2. Sparks across the gap of the spherical electrodes produce EMR.
    3. EMR induces an oscillating current in the loop.
    4. The resulting charge separation across the gap in the loop produces sparks.
- When the two gaps of the electrodes and the loop were held perpendicular to each other, there was no sparking.
    - This is because there is no charge separation occurring.





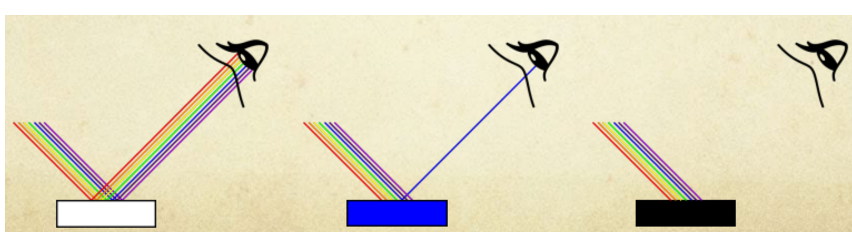


- The electromagnetic waves that Hertz produced between the spherical electrodes were known as 'Hertzian waves', now known as radio waves.
- Hertz confirmed many properties of EMR, including reflection, refraction, focusing, polarisation and that they travelled at the speed of light.

Quantum Model

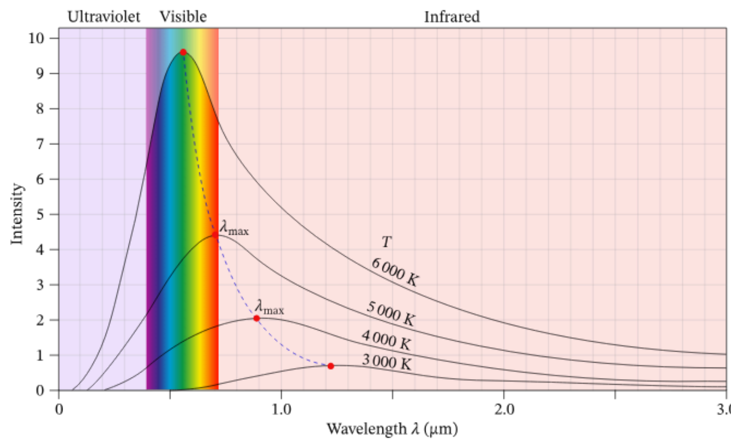
Black Body Radiation

- Colour is observed due to the reflection of light from the surface of the object.
    - White objects reflect all colours of visible light.
    - Coloured objects absorb some colours, but reflect other colours, which give them their colour.
    - Black objects absorb almost all colours of light.

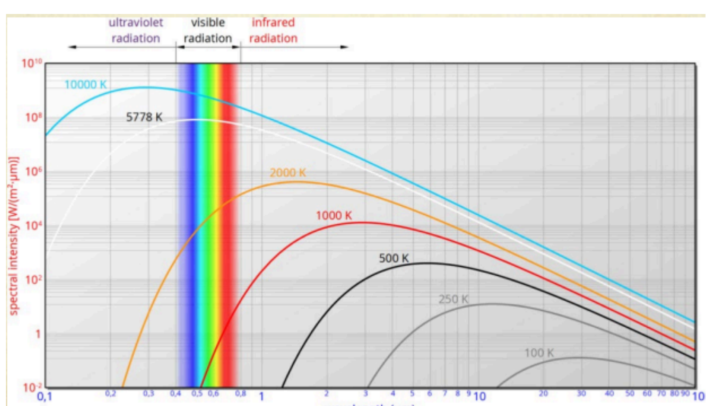


- A black body is an idealised physical object that absorbs all incident EMR, regardless of frequency or angle of incidence.

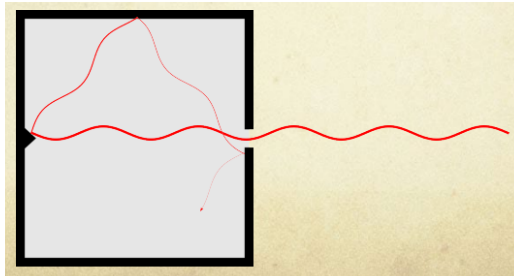
- However, they do not necessarily appear black.
    - This is because black bodies can emit EMR based on their temperature.
- As an object heats up, it changes colour because of changes in the EMR being emitted by the object.



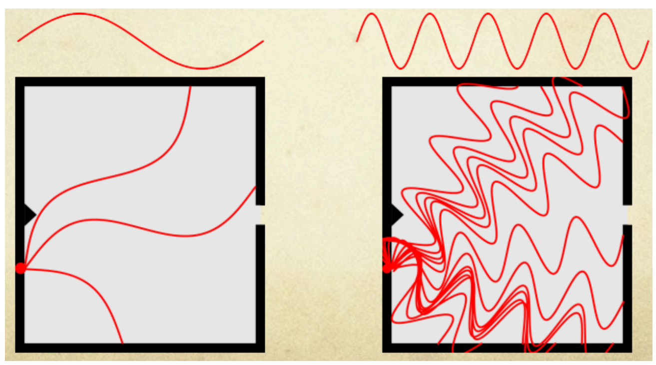
- Experiments found that the intensity of radiation increases to a peak as wavelength decreases, and drops to zero at shorter wavelengths.
- Wilhelm Wien introduced his displacement law ($\lambda_{max} = \frac{b}{T}$) in 1893, where λ_{max} is the peak wavelength (m), b is Wien's displacement constant and T is the temperature (K).



- A black body can be approximated by a cavity with black coating inside.
    - The EMR enters through the opening and reflects until it is all absorbed.



- Lord Rayleigh and James Jeans derived a classical law for black body radiation.
  - They considered the number of wavelengths that can satisfy zero displacement conditions at the walls of the cavity.

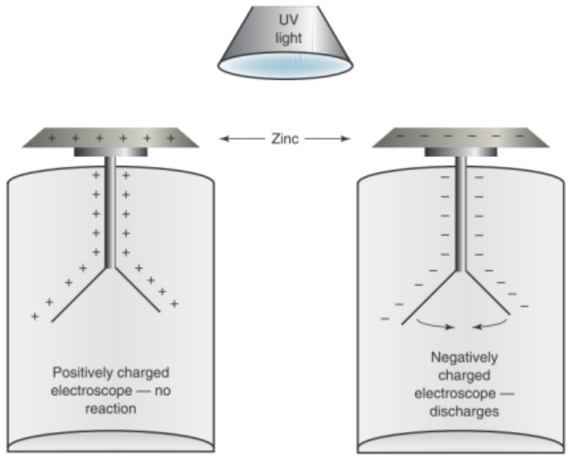


- In 1900, they derived that intensity was inversely proportional to wavelength in the relationship $I \propto \frac{1}{\lambda^4}$.
  - They later derived a formula $I = \frac{2ckT}{\lambda^4}$, where k is the Boltzmann constant.
- However, this theory implies that the radiation intensity increases dramatically as the wavelength approaches zero.
  - This violates the Law of Conservation of Energy.
- This discrepancy to experimental values was later termed the ultraviolet catastrophe in 1911, due to its inaccuracy at shorter wavelengths.
  - It was named after ultraviolet EMR because it was the only known form of EMR known to have a shorter wavelength than visible light.
- Max Planck recognised two factors with respect to the black body radiation curve.

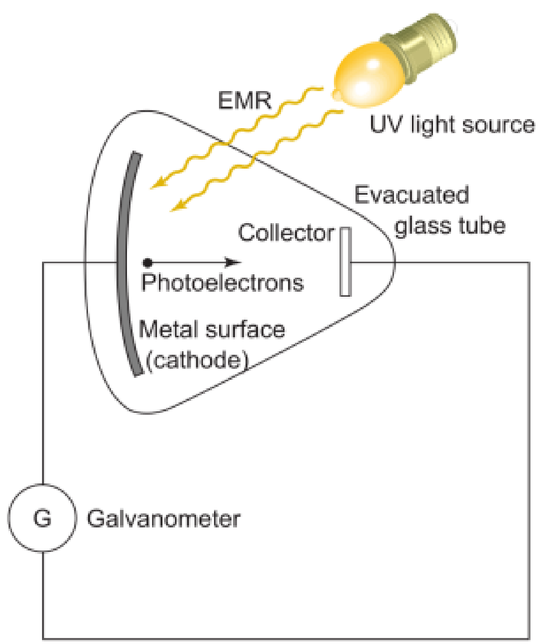
    - Rayleigh-Jeans Law was accurate at long wavelengths and Wien's Radiation Law was accurate at short wavelengths.
- In October 1900, Planck derived a preliminary formula to combine the laws.
    - He presented his new formula to the German Physical Society in December 1900.
- Planck considered that a black body had 'oscillators' of some kind on its surface to release energy.
    - He theorised that these oscillators could only vibrate at integral numbers of a fundamental frequency.
- The integral numbers of fundamental frequency lead to the quantisation of radiation energy.
    - It introduced energy elements, $E = hf$, where h is the Planck constant.
    - Planck regarded the energy as 'made up of a completely determinate number of finite equal parts'.
- Planck played a part in initiating a new branch of physics called quantum physics.
    - This branch of physics applies to very small scales such as molecular, atomic or even smaller scales.

The Photoelectric Effect

- The photoelectric effect is the release of electrons from a metal surface when exposed to EMR of sufficient energy.
- This was first observed by Hertz in his original radio waves experiments.
    - This was noticed when the spark length possible in the loop was increased when the receiver loop was illuminated with ultraviolet light.
- In 1888, Wilhelm Hallwachs designed a simpler demonstration of the photoelectric effect.

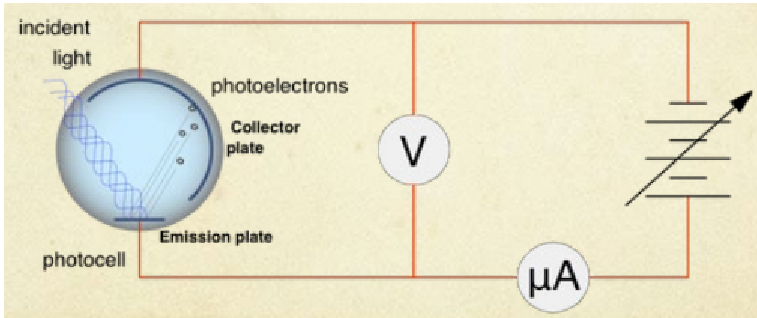


- An electroscope is used to detect the presence and magnitude of the electric charge on a body.
- In 1899, Thomson performed an experiment in a vacuum tube, where ultraviolet light was incident on the metal cathode.
    - A current was detected in the galvanometer and photoelectrons were found to have the same properties as electrons.

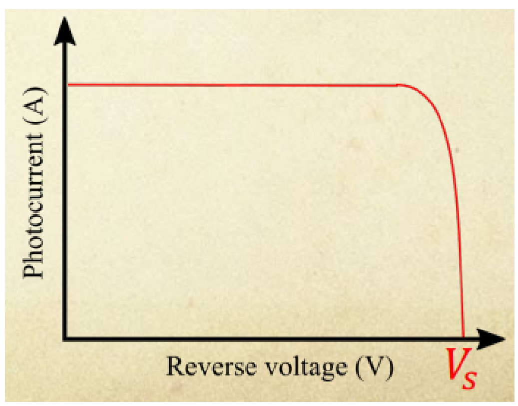


- Philipp Lenard identified a method to measure the kinetic energy of photoelectrons.

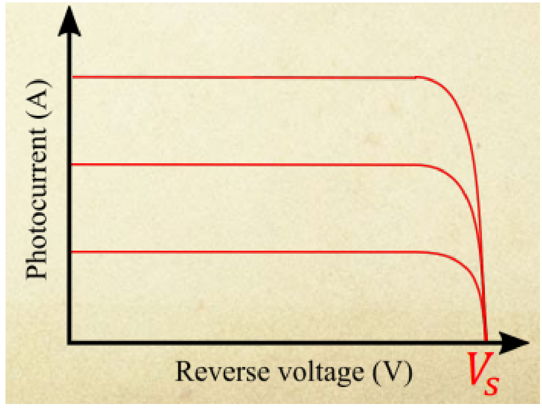
    - Using the fact that an electric field has a force on electrons, he applied a reverse voltage to accelerate electrons back into the emission plate.



- Lenard increased the reverse voltage applied and observed that the photocurrent remained constant, and then quickly dropped to zero at a certain voltage.
    - This voltage corresponded to an electric field strength sufficient to stop all photoelectrons from reaching the collector plate.
    - This is called the stopping voltage.



- Lenard repeated the same experiment with different intensities from the same light source in 1902.
    - He observed that the lower the intensity, the lower the photocurrent was.
    - The intensity did not affect the stopping voltage, so the photoelectrons had the same kinetic energy, no matter the intensity of light.



- Lenard proposed that kinetic energy is inherent in the motion of electrons due to thermal energy, and electron emission from the surface was triggered by light.
    - This was the dominant theory until 1911, because it was found that the kinetic energy of photoelectrons was not affected by temperature.
    - This theory also could not properly explain the ionisation of gases.
- In 1905, Einstein proposed his quantum nature of light.
    - He said that light is made up of 'energy quanta localised at points of space that move without dividing, and can be absorbed or generated only as complete units'.
    - Work must be done to remove electrons from the surface, and if the energy of light is sufficient, then the electrons will be emitted, no matter the intensity.
    - He derived a formula for the maximum kinetic energy of photoelectrons, $K_{\text{max}} = hf - \phi$, where ϕ is the work function of the metal.
- Einstein's light quanta behaves like particles, where it must transfer all or none of its energy to the electron.
    - This explains the photoelectric effect, the ionisation of gases and black body radiation.
    - It is now a physical foundation for Planck's quantisation of energy.
- This meant that light has both wave and particle features, known as wave-particle duality.

    - Einstein's light quantum particle was named a photon in 1926 by Gilbert Lewis.
- The phenomenon of the photoelectric effect was explained, as incident light knocks electrons from the surface atoms of the metal.
- Kinetic energy is usually expressed in units of joules, but due to the small energies involved with photoelectrons the electronvolt (eV) is used.
    - The electronvolt is the work done by accelerating an electron by a potential difference of $1V$.
    - $1\ eV = 1.602 \times 10^{-19} J$.

Special Relativity

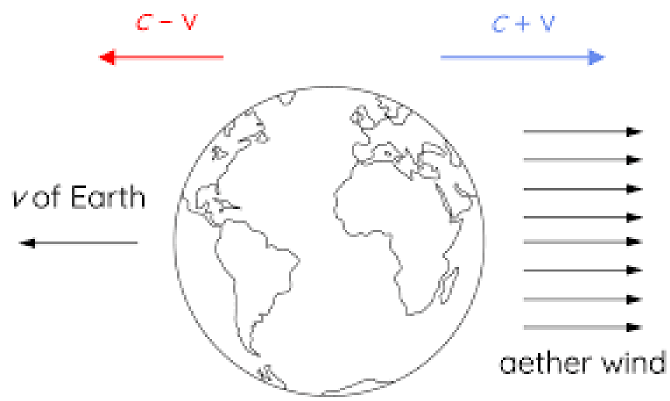
Inertial and Non-Inertial Frames of Reference

- Galileo proposed an experiment in 1632, where a ship is standing still with insects and a fish bowl below deck.
    - If the ship moves with constant velocity, then the motion of the insects, the fish, and the water inside the fishbowl will be identical to if the ship was still.
    - This meant that it could not be determined if the ship was moving without observing the moving landscape outside.
- Classical relativity applied Galileo's experiment to Newtonian mechanics, where the ultimate frame of reference is absolute space.
    - Newton's Laws of Motion hold in all frames of reference in relative uniform motion to absolute space
    - All frames of reference in relative uniform motion to absolute space share a universal time.
- An inertial frame of reference is a frame of reference that has a constant relative velocity, which can be zero.
    - All laws of physics are always valid in this frame of reference.

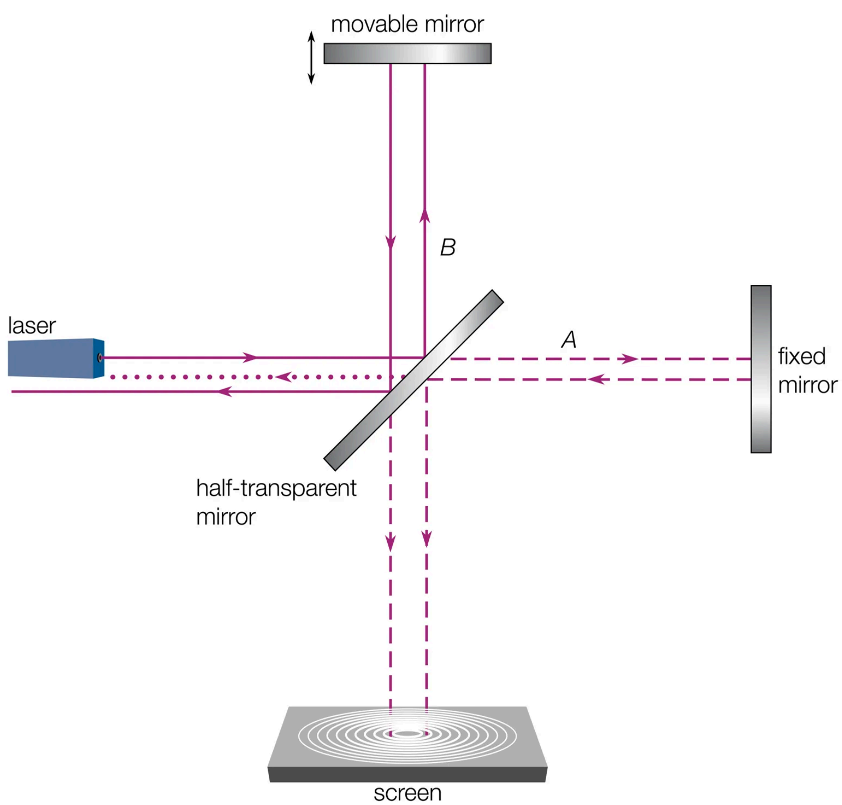
    - An experiment can not be performed to determine whether the frame of reference is stationary or moving with a constant velocity.
- A non-inertial frame of reference is a frame of reference that undergoes acceleration, including circular motion.
    - The laws of physics do not hold in this frame of reference, and fictitious forces are required to explain motion.
- The Earth appears to be an inertial frame of reference, but is actually non-inertial due to the surface of the Earth undergoing acceleration.
    - This is because of the rotation of the Earth, the Earth's orbit around the sun, and the sun's orbit around the galaxy.

The Aether and the Michelson-Morley Experiment

- The theory of light in the late 19^{th} century had a wave nature.
    - This meant that light could refract, diffract, undergo interference and be polarised.
    - However, it was believed all waves propagate in a medium and transverse waves such as light needed a solid medium.
- The luminiferous aether was postulated to explain light reaching to the Earth from the sun.
    - This aether was stationary in space, completely filled all of space, permeated all matter, was perfectly transparent, had low density, and high elasticity.
    - This elasticity was described to have solid behaviour with instantaneous changes such as light, but slower, constant changes such as the movement of planets could be accommodated.
- Scientists suggested that the Earth moving through the aether would produce an 'aether wind' in the opposite direction.
    - This aether wind is visualised below:



- The Michelson-Morley experiment planned to use the interference of light to detect and measure the relative velocity of Earth through the aether.
    - Albert Michelson developed the first interferometer in 1881, but it was not sensitive enough to perform the experiment.
    - Michelson then collaborated with Edward Morley in 1887 to develop a more precise interferometer.
- The Michelson-Morley experiment was set up as shown:



- The Michelson-Morley experiment observed that no change in the interference pattern was observed, within error of 4-8 km/s.
    - This experiment was repeated many times throughout the year, and still had a null result.
- The Michelson-Morley experiment was repeated by other scientists with reduced error.
    - Morley and Miller (1902-4) with an error of 3.5 km/s.
    - Joos (1903) with an error of 1.5 km/s.
    - Cedarholm et al. (1958) with an error of 30 m/s.
    - Trimmer et al. (1973) with an error of 2.5 cm/s.
- Michelson-Morley's results were met with scepticism as scientists wished to hold onto the aether model, and they proposed reasons for the null result.
    - 'Aether dragging' was postulated, where the aether was dragged along with the Earth's motion, removing any relative motion at the surface of the Earth.
- It was also postulated that objects contracted along the direction of motion, and Hendrik Lorentz derived length contraction according to the formula:

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}}$$

Einstein's Postulates

- Albert Einstein's fundamental premise was that the laws of physics were the same in all inertial frames of reference, including the laws of electromagnetism.
    - Einstein was aware of the Michelson-Morley experiment. but has denied it as a major influence.
- Einstein developed his theory through a series of thought experiments.
    - This was due to the effects of his theory only being detected at relativistic speeds and therefore being impossible to experimentally achieve at the time.
- In 1905, Einstein published his ideas on relativity in a paper called 'On the Electrodynamics of Moving Bodies'.

    - The paper identified two fundamental principles; the Principle of Relativity and the Principle of the Constancy of Light.
- The Principle of Relativity states that the laws of physics are the same in all inertial frames of reference.
- The Principle of the Constancy of Light states that the speed of light is c in all inertial frames of reference.
- This theory came to be known as Special Relativity.
- Einstein derived all results of previous experiments without the need for ad hoc hypotheses.
    - This meant that there was no longer any need for the aether model.
- Einstein stated that if c is constant according to $c = \frac{distance}{time}$, then two observers moving at different speeds will measure different distances and times to calculate c.
    - This meant that space and time were relative.
- Hermann Minkowski expanded on this to introduce the concept of Minkowski space, or space-time.
- This meant that in the early 20th century, there were two main theories on the nature of light.
    - The aether model stated that light travelled at c in luminiferous aether, which acted as an absolute frame of reference.
        - The measured speed of light depends on the velocity of the observer's frame of reference.
        - Space and time are constants and motion is defined by them.
    - Special relativity stated that all inertial frames of reference are equivalent and the speed of light in a vacuum is c in all inertial frames of reference.
        - The measured speed of light is constant and space and time change to accommodate this.
- If the speed of light is c in all inertial frames of reference and space and time are relative, then time dilation, length contraction and relativistic momentum occur as effects.

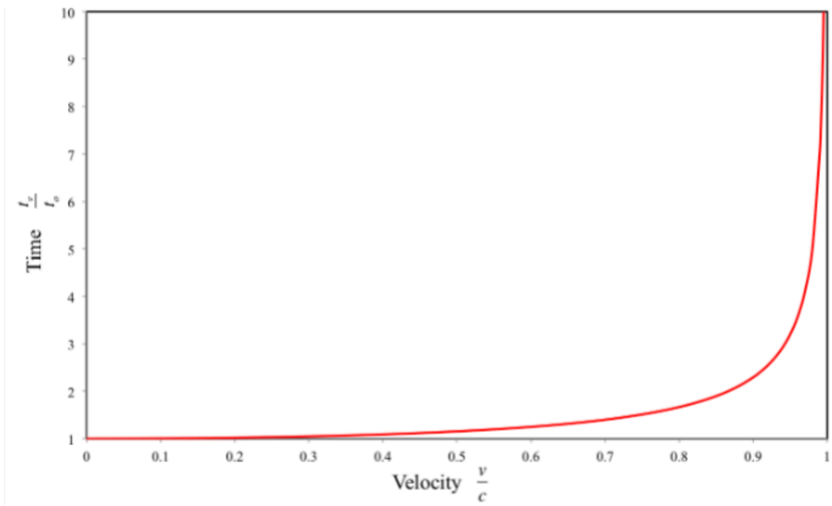
- Although not a consequence, mass-energy equivalence is related to this.

Time Dilation and Length Contraction

- The time dilation formula is below:
  - t_o is the time in an inertial frame of reference at rest with respect to the observer and t_v is the time in an inertial frame of reference that is moving with respect to the observer.

$$tv = \frac{t0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

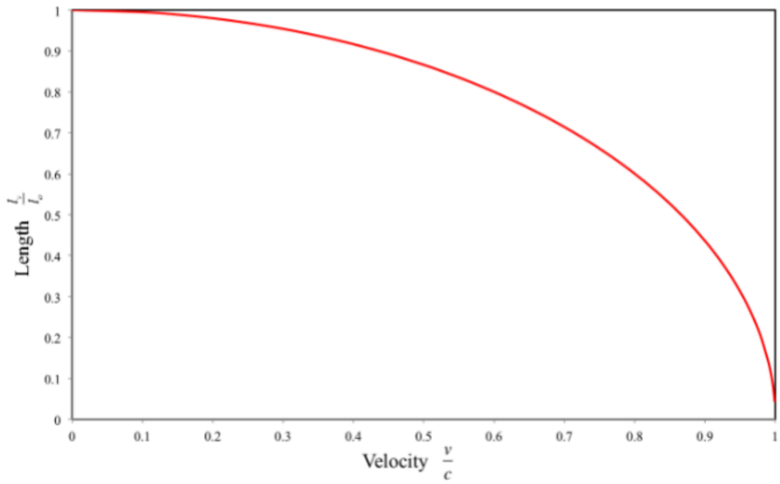
- A graph of velocity $\frac{v}{c}$ against time $\frac{tv}{t0}$ is shown below:



- The length contraction formula is below:
  - l_0 is the length in an inertial frame of reference at rest with respect to the observer and l_v is the length in an inertial frame of reference that is moving with respect to the observer.

$$lv = l0\sqrt{1 - \frac{v^2}{c^2}}$$

- A graph of velocity $\frac{v}{c}$ against length $\frac{lv}{l0}$ is shown below:



- Due to the discovery of the relative nature of space and time, the standardisation of length has been affected by relativity.
- The definition of the metre has changed over time.
  - In 1793, a metre was defined as one ten-millionth of the distance from the north pole to the equator, passing through Paris.
  - In 1960, a metre was defined as 1 650 763.73 wavelengths of orange-red emission spectrum line of the krypton-86 atom in a vacuum.
- However, both these definitions rely on physical length, and relativity calls for a definition in terms of a universal constant that is the same for all frames of reference.
- The new definition of the metre was made in 1983, which is defined as the length of path travelled by light in vacuum during a time interval of $\frac{1}{299\,792\,458}$ of a second.

Evidence for Special Relativity

- Evidence for special relativity was experimentally found, and the major evidence stems from:
  - Observations of cosmic-origin muons at the Earth's surface.
  - Time dilation measured by atomic clocks in the Hafele-Keating experiment.
  - Evidence from particle accelerators.

Cosmic-Origin Muons

- Experiments with muons were conducted in the Rossi-Hall experiment in 1941, but more accurate observations were made in the Frisch-Smith experiment in 1963.
  - Muons were stopped in a scintillator at the top of Mt Washington, and the number of muons counted in an hour and the decay times were measured.
  - This was also done at the bottom of the mountain, and they compared the number measured to the prediction made through the decay times and the time it would take for the muons to reach the bottom of the mountain.
- There was a significantly higher number of muons observed at the bottom of the mountain than hypothesised, and time dilation was shown to have extended their apparent lifespan.

Hafele-Keating Experiment

- Hafele and Keating took four highly accurate atomic clocks around the world on an airplane twice, going both east and then west.
- The time on the clocks were then compared to atomic clocks at the United States Naval Observatory, and the time lost and gain was recorded.

	Predicted Time Gained (ns)			**Measured Time Gained (ns)**
	General Relativity	**Special Relativity**	**Total**	
Eastward	+144±14	-184±18	-40±23	-59±10
Westward	+176±18	+96±10	+275±21	+277±7

- The measured time was extremely close to the predicted time hypothesised due to time dilation.
  - However, general relativity is needed to fully explain the measured time in comparison to the prediction.

Evidence from Particle Accelerators

- Particle accelerators can achieve relativistic speeds for small particles.
  - These speeds can reach up to 0.999999999988c.
- Scientists measured the rate of transitions between energy states in stationary atoms, and measured the same transitions at one-third the speed of light in a particle accelerator.
- The values of the transition rates agreed with Special Relativity.

Adding Velocities

- Consider a train travelling at 0.75c with respect to an observer and a tennis ball is thrown forwards at 0.75c with respect to the train.
  - The tennis ball is measured at 0.75c from the train, but lengths contract and time slows down with respect to the observer, making the actual speed of the ball different to what's observed on the train.
- The speed of the tennis ball is 0.96c, calculated through the formula:

$$w = \frac{u+v}{1+\frac{uv}{c^2}}$$

Relativistic Momentum

- When calculating momentum at relativistic speeds using Newtonian physics, momentum is not conserved during collisions.
- This means that relativistic momentum occurs, which is the momentum of an object travelling at relativistic speed.
- The relativistic momentum formula is below:
  - p_v is the momentum in an inertial frame of reference that is moving with respect to the observer and m_0 is the mass of the object at rest.

$$pv = \frac{m0v}{\sqrt{1-\frac{v^2}{c^2}}}$$

- Applying a force to an object increases its speed, and thus also increases its kinetic energy.

- It also increases the object's momentum, meaning that mass associated with momentum appears to increase.
  - This hints towards a relationship between mass and energy.
- An object's speed is limited by special relativity.
  - This is because as an object with mass approaches the speed of light, the relativistic momentum of the ship approaches infinity.
  - According to the formula $F = \frac{\Delta p}{t}$, an infinite force would be required for the object to reach the speed of light, which is not possible.
  - The exception to this speed limit is the speed of massless particles, such as photons.

Mass-Energy Equivalence

- By defining a relativistic mass through the relativistic momentum formula, rearranging the formula relates energy and mass:
  - m_v is the apparent mass of the object that is moving with respect to the observer and m_0 is the mass of the object at rest.

$$mv = \frac{m0}{\sqrt{1-\frac{v^2}{c^2}}}$$

- The relativistic increase in mass multiplied by c^2 was found to be equal to the kinetic energy.
- Energy is thus 'converted' into mass and mass and energy are considered to be equivalent according to:

$$E = mc^2$$

Space Travel and the Twins Paradox

- If a rocket were to travel at relativistic speeds, the following effects of special relativity would occur:
  - The distance to the destination decreases due to length contraction from the rocket's frame of reference.

    - The time experienced by passengers decreases due to time dilation from Earth's frame of reference.
    - The rocket's ability to accelerate is limited due to relativistic momentum and mass-energy equivalence.
- The Twins Paradox is detailed through the following hypothetical situation:
    - Consider twins A and B, where A boards a rocket and travels close to c to a distant star and returns at the same speed, while B stays on Earth.
    - Due to time dilation, A experienced less time and returned to Earth younger than B, but from A's perspective, B on Earth moves at relativistic speed and should be younger.
- The Twins Paradox assumes that both frames of reference are equivalent, where B is in the inertial frame of reference of Earth and A's frame of reference is only inertial when travelling at constant speed.
    - When accelerating and decelerating, A is then in a non-inertial frame of reference.
- The Twins Paradox is fully resolved in General Relativity.